

Balancing Signal and Noise: Context Saturation in Multimodal Large Language Models for Misinformation Detection

Anonymous ACL submission

Abstract

[Background:] Large Language Models (LLMs) are increasingly used for misinformation detection, where contextual information plays a critical role in model decision-making. [Problems & Research Gap:] Although prior studies show that context can improve LLM-based misinformation detection, it remains unclear how much context is actually beneficial and at what point additional information becomes harmful, especially when the context contains noisy, irrelevant, or misleading evidence.

[Main contribution 1:] In this paper, we investigate the relationship between context quantity and misinformation detection performance in LLMs. We introduce the concept of the **Context Saturation Effect**, which describes a phenomenon where adding contextual information improves performance only up to a certain threshold. Beyond this point, excessive or misleading context no longer enhances reasoning and instead degrades model reliability, increasing the likelihood of misclassification.

[Main contribution 2:] To address this issue, we propose an **adaptive context selection framework** that dynamically balances informative signals and noisy evidence during retrieval. Rather than simply increasing the amount of retrieved context, our method aims to identify and prioritise high-quality contextual information while suppressing irrelevant or misleading content. Experimental results are expected to provide new insights into how context influences LLM reasoning in misinformation detection and how adaptive retrieval strategies can improve robustness and reliability.

[Findings:] We conduct controlled experiments across multiple context levels and context types, including relevant, ambiguous, and misleading information, using both vanilla LLM prompting and retrieval-augmented setups. Our results reveal a consistent non-linear pattern: moderate amounts of high-quality context yield the

best performance, while excessive or noisy context leads to increased misclassification, particularly under misleading conditions.

[Further analysis] shows that LLMs are sensitive not only to context length but also to context quality, with misleading context disproportionately affecting false positive and false negative rates.

[Impact:] the importance of balancing signal and noise in context design and provide practical insights for prompt engineering, retrieval strategies, and robust misinformation detection.

keywords: misinformation detection, large language models, retrieval-augmented generation, context-aware reasoning, evidence selection, context saturation, noisy retrieval, multimodal misinformation robustness, hallucination mitigation

1 Introduction

We define **context** as any auxiliary evidence retrieved or associated with multimodal content that may influence LLM reasoning, including textual descriptions, metadata, social interactions, external evidence, and cross-modal relationships. [background:] 1. Growth of LLMs in misinformation detection. 2. Role of retrieval and contextual grounding. 3. Multimodal and social-context misinformation challenges.

[motivation]: Humans also suffer from information overload. 1. A statement can seem believable or correct when you only see the statement itself, but once additional background information is provided, you may realise it is misleading, incomplete, or false. 2. Correct context helps 3. Too much irrelevant or conflicting context confuses the model.

[Problem Statement]: How much context is beneficial? When does context become harmful?

[Our findings]: RQ1. How does context quantity affect LLM misinformation detection performance?

084	RQ2.How does context quality influence model reliability?	3.3 Visualization of Context Saturation	126
085		Figure 1: Context Saturation Curve.	127
086	RQ3.Do different LLMs exhibit different saturation thresholds?	Figure 2: Performance under varying noise ratios.	128
087		Figure 3: Attention distribution across context lengths	129
088	RQ4.Can adaptive context selection mitigate context saturation?		130
089			131
090	[Contributions]: 1. threshold effect .2. noise accumulation. 3. retrieval overload; 4. performance degradation; 5 adaptive balancing		
091			
092			
093	2 Related work	4 Adaptive Context Selection	132
094	2.1 LLM-based Misinformation Detection	Novel method: Stop retrieval before saturation	133
095	[Defines application domain] 1. Prompting. 2. RAG-based verification. 3. Multimodal misinformation detection	Possible components: [1]Relevance Scoring ;[2]Measure semantic alignment; [3]Noise Detection; [4]Detecting misleading or low-quality evidence ;[5] Dynamic Context Filtering	134
096			135
097			136
098	2.2 Information overload		137
099	[Provides theoretical motivation] Cognitive overload Irrelevant evidence Distractor information Hallucination amplification	5 Experiments	138
100		5.1 Task	139
101		Input: Claim/content Retrieved contextual evidence Output: True / false / Explanation	140
102	2.3 Context length studies		141
103	[LLMs struggle with long or noisy contexts]	5.2 Datasets	142
104	Long-context reasoning Lost-in-the-middle problem Context dilution	5.3 Context category	143
105		5.4 Context Construction	144
106	2.4 Context-aware retrieval	Context Quantity Levels: 0 context ;1 evidence ;3 evidence ;5 evidence; 10 evidence;Full retrieval ;Context Quality Types;Relevant Context	145
107	[Defines technical background] Retrieval-Augmented Generation (RAG) Evidence selection Context ranking Knowledge grounding	Supports accurate reasoning: Ambiguous Context; Weakly related; Misleading Context;Contradictory/noisy evidence ;Redundant Context	146
108			147
109			148
110	2.5 Research Gap		149
111	Better retrieval Larger context windows But lacks: Understanding optimal context threshold Saturation behaviour Noise-context interaction	5.5 Models	150
112		Models: OpenAI GPT-4o ,Anthropic Claude , Google Gemini, Llama 3, Qwen2.5	151
113			152
114	3 Context Saturation Effect		153
115	3.1 More context is not always better	5.6 Prompting	154
116	Context Saturation Effect means Adding contextual information helps an LLM only up to a certain point. Beyond that point, additional context no longer improves performance and may instead reduce reasoning reliability due to noise, distraction, redundancy, or misleading evidence.	Vanilla prompting, Chain-of-thought, RAG prompting	155
117			156
118		5.7 Evaluation	157
119		5.8 Results	158
120		5.8.1 Context Saturation Effect	159
121		Possible figure: X-axis = context amount Y-axis = performance Performance rises then declines. Could show: Optimal zone , Noise-dominant zone	160
122	3.2 Why Context Saturation happens		161
123	[Attention dilution], [Contradictory evidence], [Retrieval redundancy], [Token competition], [Reasoning distraction],		162
124			163
125			

5.8.2 Adaptive Context Selection

5.9 Findings

5.9.1 RQ1.How does context quantity affect LLM misinformation detection performance?

5.9.2 RQ2.How does context quality influence model reliability?

5.9.3 RQ3.Do different LLMs exhibit different saturation thresholds?

5.9.4 RQ4.Can adaptive context selection mitigate context saturation?
Performance curves, Saturation points, Degradation regions

5.10 Implications and future research

5.10.1 Why Saturation Happens

5.10.2 Implications for RAG

5.10.3 Implications for Misinformation Detection Trustworthiness Robust reasoning Evidence reliability

5.10.4 Broader Implications

Could generalise beyond misinformation, for example, Medical reasoning Legal AI

5.11 Limitations

For Examples: Dataset bias, Retrieval dependency, Closed-source model constraints, English-only evaluation

5.12 Conclusion

Context helps only to a point Excessive context harms reasoning, Adaptive selection improves robustness